

Step 2: Exploring Alternatives

Alternative 1:

Use a train sensor with the vehicle detection system. The gates also lower if a vehicle is detected to be on the tracks. The barriers only rise once the train has passed and no vehicle is on the tracks.

Alternative 2:

Implement predictive timing and continually improve the model. The vehicle sensors additionally check whether the tracks are clear before opening the gates.

- **Real-World Example:**

In Australia, most railway level crossings apply a system of track circuits and/or axle counters for train detection (RISSB, 2020). The system engages the flashing lights and warning bells before lowering the gates. The systems also incorporate sensors that detect the presence of a vehicle or pedestrian to ensure the gates do not lift prematurely.

Some pedestrian crossings are equipped with CCTV cameras to improve safety and monitor activity. Fail-safe design principles ensure that gates remain closed in the event of a system failure. The method is consistent with safety regulations in the railway industry, which always come first.

Steps 3 and 4: Evaluating Alternatives + Engineering Decision

- **Evaluation:**

Alternative 1 is simple, reliable, and ensures gates are always lowered if a hazard is present.

Alternative 2 adds predictive timing but increases complexity and the chance of errors.

- **Engineering Decision**

I will use **Alternative 1** as it is safer, reliable and easier to implement. It directly addresses the two main risks: an approaching train and a vehicle stuck on the tracks. Its simplicity reduces the risk of malfunction and increases reliability.